

Development of an Optimized Biogas Storage Solution with Consideration of Dual Application in CAES

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Introduction

This report is a chronological documentation of an attempt at development of a biogas storage solution which is suited for use in a decentralized system and may be used as a CAES vessel when the fuel is exhausted. Initial stages of the project involved simultaneous research into biogas fundamentals and compressed air energy storage (CAES) systems. Following this, a rigid pressure vessel approach was selected as the most feasible due to advantages in portability, structural reliability, and potential future extensibility as a CAES vessel.

Further investigation into isochoric and isobaric CAES systems was subsequently conducted to evaluate the possible integration of CAES-inspired operation into the proposed vessel design. However, continued exploration of complete CAES implementation resulted in unforeseen expansion of project scope due to the increasing complexity of thermodynamic modeling, pressure regulation, and compressor-expander integration. As a result, the project focus was narrowed primarily toward pressure vessel engineering and storage optimization, while retaining limited CAES-related calculations for comparative purposes. Following this transition, the work branched into two parallel directions consisting of adsorption-based storage research and ASME-based pressure vessel design methodology.

A feasibility investigation into high-pressure biogas storage was then performed to evaluate the practicality of moderate-pressure operation under realistic fabrication and safety constraints. Benchmarking studies based on a domestic 14.2 kg LPG cylinder were car-

ried out to establish practical energy storage targets and deployment constraints. Based on these studies, a target equivalent energy storage of approximately 25 to 35% relative to a standard LPG cylinder was selected, motivated by considerations of weekly refill practicality. Additional calculations relating biogas energy density to vessel pressure and volume were performed in order to identify the relationship between storage capability and structural requirements.

Subsequent manual optimization attempts involving pressure-volume tradeoffs and material variation revealed significant computational and practical limitations. Although several conceptual sets of parameters for the shell itself were generated, all were found to be impractical, either due to extreme volume or weight constraints impairing portability. This resulted in the introduction of both mass and vessel size constraints in future optimizations.

Due to the increasing complexity of simultaneously satisfying structural, geometric, energetic, and practical constraints, purely manual analytical optimization approaches became impractical. Consequently, the project transitioned toward the development of an algorithmic optimization framework in MATLAB using constrained cost functions and identified design variables. The present work therefore serves not as a finalized vessel design, but as a structured engineering framework documenting the evolution of the project from broad exploratory research toward constrained computational optimization of a moderate-pressure biogas storage vessel.

Contents

1 Preliminary Exploration of Biogas, CAES, and Existing Storage Systems	5
1.1 Biogas Fundamentals	5
1.1.1 Composition of Biogas	5
1.1.2 Identified Challenges and Preliminary Assumptions	5
1.2 Introduction to Compressed Air Energy Storage (CAES)	6
1.3 Biogas Compression Requirements and Equipment	7
1.3.1 The Necessity of Compression	7
1.3.2 Example Compression Apparatus	7
1.3.3 Feasible Pressure Range For Storage	8
1.4 Final Outcome of Preliminary Exploration	8
2 Branching of Research Directions	10
2.1 Adsorption-Based Storage Research	10
2.1.1 Motivation for Adsorption Storage	10
2.1.2 Principles of Physical Adsorption and Biochar	10
2.1.3 Experimental Results (Khan et al., 2022)	11
2.1.4 Conclusion on Adsorbed Biogas Viability	11
2.2 Pressure Vessel Design Research	12
2.2.1 Introduction to ASME Manuals	12
3 Energy Density Benchmarking	13
3.1 Selection of LPG Cylinder as Benchmark System	13
3.2 Energy Density Calculations for LPG Storage	13
3.3 Establishment of Biogas Storage Target	13
3.4 Rationale Behind the Selected Target	13
3.5 Calculation of Biogas Energy Density as a Function of Pressure and Volume	14
3.5.1 Need for Pressure-Based Energy Density Estimation	14
3.5.2 Ideal Gas Approximation	14
3.5.3 Methane Fraction Considerations	14
3.5.4 Stored Chemical Energy Estimation	15
3.5.5 Volumetric Energy Density Relation	15
3.5.6 Pressure-Volume Tradeoff Analysis	15
3.5.7 Implications for Vessel Design	15
4 Manual Optimization Attempts	17
4.1 Manual Design Exploration	17
4.2 Practical Difficulties Encountered	17

4.3	Impracticalities Observed in Preliminary Designs	18
4.4	Results	18
5	Algorithmic Optimization Attempt	19
5.1	Need for Computational Optimization	19
5.2	Definition of the Design Problem	20
5.3	Research References and Methodology Inspiration	21
5.4	Selection of Optimization Variables	21
5.5	Constraint Formulation	22
5.6	Objective Function Formulation	23
5.7	MATLAB-Based Optimization Framework	24
5.8	Material Dataset and Property Mapping	24
5.9	Post-Optimization CAD and FEA Validation	25
5.10	Proposed Improvements to the Optimization Pipeline	26
6	Conclusion	28

1 Preliminary Exploration of Biogas, CAES, and Existing Storage Systems

1.1 Biogas Fundamentals

Biogas is a gaseous fuel generated primarily through the anaerobic digestion of organic matter. Its composition depends strongly on the substrate being digested, digestion conditions, and purification processes performed after production. Raw biogas typically consists primarily of methane and carbon dioxide, along with trace quantities of water vapour, oxygen, sulphur compounds, and other impurities. Due to the combustible nature of methane, biogas presents potential as a decentralized renewable fuel source, although its low volumetric energy density under atmospheric conditions significantly limits storage and transport practicality at atmospheric pressure.

1.1.1 Composition of Biogas

The composition of biogas displays variance with change in the substrate being digested, but the following table, which has been compiled after identifying the upper and lower values of the percentages for various substrates, can be used as a reference for approximate composition.

Table 1: Typical Composition of Raw Biogas

Component	Typical Composition
Methane (CH_4)	50 to 75%
Carbon Dioxide (CO_2)	25 to 50%
Water Vapour	Trace
Oxygen	$\approx 0.5\%$
Sulphur Compounds	0 to 3%

1.1.2 Identified Challenges and Preliminary Assumptions

During the preliminary investigation phase, several technical and practical challenges associated with compressed biogas storage were identified.

- **Low Volumetric Energy Density**

Biogas under atmospheric conditions possesses relatively low volumetric energy density, making direct low-pressure storage impractical for portable decentralized applications. This resulted in the conclusion that gas compression would be necessary in order to obtain usable storage capacity within a realistically sized vessel.

- **Variable Biogas Composition**

Biogas composition varies depending on substrate type, digestion conditions, and plant operation. This variability introduces uncertainty in energy density calculations and storage behaviour. To simplify subsequent calculations and optimization, later stages of the project assumed that sufficiently purified biogas or biomethane-like gas would be supplied to the vessel.

- **Methane Leakage**

Methane possesses significantly higher greenhouse impact than carbon dioxide, making leakage prevention an important design consideration. This resulted in early emphasis on sealing reliability and leakage minimization.

- **Air Ingress and Combustion Hazard**

The introduction of oxygen into a compressed methane-rich environment may create explosive conditions, further emphasizing need for sealing reliability.

- **Microbial Contamination**

Raw biogas may contain microorganisms originating from anaerobic digestion processes. This issue was assumed to be mitigated through prior purification and gas treatment before storage.

1.2 Introduction to Compressed Air Energy Storage (CAES)

Compressed Air Energy Storage (CAES) is a method of energy storage in which gas is compressed using external energy input and later expanded to recover useful work or power [2, 4]. Conventional CAES systems are typically used for large-scale energy storage applications, where excess electrical energy is converted into stored pressure energy through gas compression.

Multiple CAES storage methods were identified during the initial research phase, including isochoric storage systems, isobaric storage systems, underground cavern storage, piston-based variable volume systems, and pressure-regulated storage arrangements [2, 3, 4, 5]. Isochoric systems store gas within a fixed-volume vessel and operate through pressure variation during charging and discharge, whereas isobaric systems attempt to maintain approximately constant pressure through moving boundaries or variable-volume chambers.

Further investigation into these systems highlighted the increasing mechanical complexity associated with variable-volume and pressure-regulated storage systems [1]. Requirements involving moving boundaries, dynamic sealing arrangements, and mechanically coupled pressure regulation mechanisms were identified as potentially impractical within the context of a small-scale decentralized storage system.

As a result, the preliminary exploration gradually narrowed toward comparatively simpler rigid-body storage approaches based primarily on fixed-volume pressure vessels, due to their reduced mechanical complexity.

1.3 Biogas Compression Requirements and Equipment

1.3.1 The Necessity of Compression

At standard atmospheric pressure and temperature, biogas possesses a critically low volumetric energy density, typically ranging from 20 to 25 MJ/m^3 depending on its methane concentration. As demonstrated in earlier thermodynamic evaluations, an uncompressed 33.3 L vessel would store less than 1 MJ of energy. This is entirely insufficient for practical decentralized applications such as domestic cooking or adequate energy recovery in a CAES system.

To bridge the gap between production volume and practical storage capacity, the gas must be mechanically compressed. Compression forces the gas molecules into a much tighter proximity, drastically increasing the energy stored per unit of internal vessel volume. Even with the integration of Adsorbed Biogas (ABG) technology, a baseline compression stage is mandatory to overcome the low atmospheric density and force the methane molecules into the microscopic pores of the activated biochar at the target working pressure of 20 bar.

1.3.2 Example Compression Apparatus

Achieving the target pressure of 20 bar requires a specialized multi-stage or high-capacity compressor. A representative example of industrial equipment capable of meeting these parameters is the CS20B1 201001 piston compressor.

This unit provides a practical benchmark for the mechanical energy input required for the proposed decentralized storage network. The key specifications of this compressor include:

- **Working Pressure:** 20 bar, which aligns precisely with the targeted moderate-pressure design for the ABG and CAES vessel applications.
- **Compressor Type:** Piston-driven, capable of taking low inlet suction pressures (0 to 0.09 bar) and stepping them up to the required storage pressure.
- **Flow Rate:** 0.34 m^3/min , determining the volumetric rate at which the storage vessels can be charged.
- **Motor Power:** 5 HP (3.7 kW) operating on a 380V supply, highlighting the electrical energy expenditure required to pressurize the system.

Understanding the operational parameters and power draw of such equipment is crucial for evaluating the overall lifecycle energy costs and the logistical feasibility of the compression stage within the proposed decentralized biogas network.

1.3.3 Feasible Pressure Range For Storage

Research of feasibility of high pressure storage was crucial to go forward with this approach, since if no source was found that confirmed the viability of high pressure biogas storage, it would be a moot point to pursue it as an avenue for optimization. Fortunately, a source was found in [9] which referenced the following table from [10] Unfortunately, the original paper,[10] could not be obtained online, so this secondary source remains without verification.

1.4 Final Outcome of Preliminary Exploration

Initial research established that direct low-pressure storage of biogas would be impractical due to its low volumetric energy density, thereby necessitating some degree of gas compression for meaningful storage capacity. Additionally, variability in raw biogas composition, the presence of corrosive impurities, and thermal effects associated with gas compression were identified as important engineering considerations during early investigation.

Parallel investigation into CAES systems identified multiple storage approaches including isochoric, isobaric, and pressure-regulated systems. Although small-scale isochoric CAES methodology initially appeared promising due to its conceptual compatibility with compressed gas storage, increasing mechanical and thermodynamic complexity and coupled compressor-expander arrangements gradually resulted in the abandonment of CAES implementation within the scope of the project. Following this reduction in scope, the research direction branched into two parallel approaches. One branch investigated unconventional lower-pressure storage enhancement methods, involving adsorption-assisted storage concepts. The other branch pursued conventional high pressure rigid vessel design, with emphasis on energy equivalents, pressure capability and optimization of basic vessel geometry and operating parameters.

Table 2: Existing Biogas Storage Methods and Operating Conditions

Purpose of Storage	Pressure (psi)	Storage Device	Material	Size (ft^3)
54cmShort and intermediate storage for on-farm use (currently used on farms for biogas storage)	< 0.1	Floating Cover	Reinforced and non-reinforced plastics, rubbers	Variable volume, usually less than one day's production
	< 2	Gas bag	Reinforced and non-reinforced plastics, rubbers	150 – 11,000
	2 – 6	Water sealed gas holder	Steel	3,500
		Weighted gas bag	Reinforced and non-reinforced plastics, rubbers	880 – 28,000
		Floating roof	Plastic, reinforced plastic	Variable volume, usually less than one day's production
24cmPossible means of storage for later or off-farm use (could be used for biomethane)	10 – 2,900	Propane or butane tanks	Steel	2,000
	> 2,900	Commercial gas cylinders	Alloy steel	350

2 Branching of Research Directions

2.1 Adsorption-Based Storage Research

2.1.1 Motivation for Adsorption Storage

Biogas, primarily composed of methane (CH_4) and carbon dioxide (CO_2), is a renewable energy source produced through anaerobic digestion of organic matter. Its low energy density at atmospheric pressure makes storage and transport a significant challenge. Conventional compressed biogas (CBG) requires pressures up to 200 bar, demanding expensive and heavy-duty infrastructure that is impractical for rural or domestic use.

Adsorbed Biogas (ABG) offers a promising alternative. By packing a storage cylinder with a microporous adsorbent material, methane molecules are held on the adsorbent surface via weak Van der Waals forces, substantially increasing the volumetric storage capacity at much lower pressures (≤ 20 bar). This makes ABG safer and more cost-effective for small-scale and rural applications.

2.1.2 Principles of Physical Adsorption and Biochar

Physical adsorption (physisorption) occurs when gas molecules (the *adsorbate*) adhere to the surface of a porous solid (the *adsorbent*) through Van der Waals interactions. This process is fully reversible, as the gas is released simply by reducing pressure, making it ideal for practical storage and discharge cycles.

In an ABG cylinder, the adsorbent is packed tightly inside the storage vessel. The key governing principle is that an increase in surface area provides more adsorption sites, resulting in a higher volume of gas stored at the same pressure.

Biochar is produced by **pyrolysis**, which involves heating biomass (e.g., coconut shell, bamboo, pigeon pea stalk) at $\sim 500^\circ\text{C}$ in an inert (nitrogen) atmosphere for approximately 2 hours. This process burns off volatile compounds and leaves behind a carbon-rich skeleton riddled with micropores, which are the active sites for methane adsorption.

The adsorption capacity of a biochar is primarily characterised by two metrics:

- **BET Surface Area (m^2/g):** Measures total available surface for adsorption. Higher is better.
- **Iodine Number (mg/g):** A quick proxy for porosity. Higher values indicate greater micropore volume.

2.1.3 Experimental Results (Khan et al., 2022)

Recent studies have evaluated the efficacy of various biochars as fillers for low-pressure ABG cylinders. Three biochar fillers were tested and compared by Khan et al. [6]. Their key properties are summarised in Table 3.

Table 3: Characterisation of different biochar fillers [6]

Property	Pigeon Pea	Bamboo	Coconut (Activated)
Iodine Number (mg/g)	106.4	87.14	920.04
BET Surface Area (m^2/g)	85.67	30.01	957.00
Total Pore Volume (cm^3/g)	0.0697	0.0272	0.5427
Mean Pore Diameter (nm)	3.259	3.634	2.266
Carbon Content (% by wt.)	70.93	74.13	85.59

Coconut activated biochar clearly dominates in surface area and pore volume, explaining its superior adsorption performance.

Experiments were conducted at room temperature ($25 \pm 3^\circ\text{C}$) using enriched biogas containing $\sim 88.8\%$ methane. The ABG cylinder had a volume of 2.831 L with a design pressure of 30 bar. Results at 20 bar are shown in Table 4.

Table 4: Enriched biogas stored at 20 bar with different fillers [6]

Filler Used	Gas Stored (g)
No adsorbent (empty cylinder)	44
Pigeon pea stalk biochar	54
Bamboo biochar	41
Coconut activated biochar	75

Coconut activated biochar achieved a **70% improvement** over an empty cylinder, storing 75 g versus 44 g at the same pressure of 20 bar. After 5 charge-discharge cycles, coconut activated biochar also showed zero residual gas, indicating complete and reversible adsorption.

2.1.4 Conclusion on Adsorbed Biogas Viability

The ABG concept demonstrates that enriched biogas can be stored significantly more efficiently at low pressures by leveraging the high surface area of activated biochar. Coconut shell activated biochar, with a BET surface area of $957 \text{ m}^2/\text{g}$, achieved a 70% improvement in storage capacity over an empty cylinder at just 20 bar. The key to successful implementation is sourcing or producing a high-surface-area adsorbent ($> 500 \text{ m}^2/\text{g}$), proper cylinder assembly, and systematic cycle testing. If validated at lab scale, this technology

holds strong potential for low-cost rural biomethane storage without high-pressure compression infrastructure, directly addressing the volumetric constraints identified in earlier sections of this report.

2.2 Pressure Vessel Design Research

2.2.1 Introduction to ASME Manuals

For the purpose of having a reliable source for parametric geometry generation formulae for pressure vessels, the ASME(American Society of Mechanical Engineers) manuals were consulted. This is due to the abundant references available within them that allow for a streamlined calculation pipeline to generate geometry of the vessel for the required design parameters. International Standards Organization (ISO) Manuals were also considered, but ultimately not chosen due to confusing structure and references.

In subsequent sections, the general workflow for calculation of parameters is as follows:

- Selection of required pressure, vessel dimensions and material, along with physical properties of the material such as maximum allowable stress from [7].
- computation of the geometrical parameters such as thickness from [8] such as :

$$\text{Shell Thickness: } t_s = \frac{PR}{SE - 0.6P} \quad (1)$$

along with any conditions on them being used from [8]

3 Energy Density Benchmarking

3.1 Selection of LPG Cylinder as Benchmark System

To establish a practical baseline for the proposed biogas storage vessel, a standard domestic Liquefied Petroleum Gas (LPG) cylinder was selected as the benchmark. The standard 14.2 kg LPG cylinder is ubiquitous in domestic settings, particularly in India. Because its dimensions, weight, and energy capacity are universally understood standards for a month’s supply of cooking fuel, it serves as an ideal baseline to evaluate the practicality and performance of alternative decentralized gas storage systems. The internal water capacity of this standard cylinder is 33.3 liters (0.0333 m^3).

3.2 Energy Density Calculations for LPG Storage

To benchmark the energy target, the total chemical energy contained within a standard LPG cylinder must be calculated. LPG has a nominal volumetric energy density of approximately 26 MJ/L.

If the 33.3 L cylinder were entirely filled, the theoretical maximum energy would be $26 \text{ MJ/L} \times 33.3 \text{ L} = 865.8 \text{ MJ}$. However, strict safety regulations dictate that pressure vessels storing liquefied gases must not be filled to 100% capacity to allow for thermal expansion. A standard fill factor of 86% is utilized, representing 86% liquid and 14% vapor headspace. Applying this practical constraint yields $0.86 \times 865.8 \text{ MJ} = 744.588 \text{ MJ}$. Therefore, a standard domestic LPG cylinder provides approximately 744.6 MJ of usable energy.

3.3 Establishment of Biogas Storage Target

Given the inherently lower energy density of biogas compared to liquefied propane and butane mixtures, matching the 744.6 MJ capacity within the same geometric footprint at moderate pressures is thermodynamically impossible. Instead, a targeted equivalent energy storage of 25% to 35% relative to the standard LPG cylinder was established for the proposed vessel.

3.4 Rationale Behind the Selected Target

This 25% to 35% target range, providing approximately 186 MJ to 260 MJ, was selected to balance the thermodynamic limitations of compressed gas with practical user convenience. If a standard LPG cylinder lasts an average household roughly four weeks, a biogas cylinder containing 25% of that energy would necessitate a weekly refill. A once a week refill cycle is considered the upper limit of acceptable logistical burden for a decentralized

domestic energy system; any less capacity would render the cylinder too inconvenient for daily cooking applications.

3.5 Calculation of Biogas Energy Density as a Function of Pressure and Volume

3.5.1 Need for Pressure-Based Energy Density Estimation

Because biogas remains a permanent gas at standard ambient temperatures and moderate pressures, unlike LPG which liquefies, its energy density is highly dependent on the degree of compression. Developing a mathematical relationship between pressure, volume, and energy is necessary to evaluate if the 25% to 35% target can be met within practical pressure limits.

3.5.2 Ideal Gas Approximation

The behavior of the compressed biogas can be approximated using the Ideal Gas Law:

$$PV = nRT \quad (2)$$

By substituting the number of moles (n) with mass divided by molar mass (m/M), the equation can be rearranged to find the density (ρ) of the gas at a given pressure and temperature:

$$\rho = \frac{P \cdot M}{R \cdot T} \quad (3)$$

For these calculations, the molar mass (M) of standard upgraded biogas is taken as 0.027 kg/mol, and the universal gas constant (R) is 8.314 J/(mol K).

3.5.3 Methane Fraction Considerations

The total energy of the biogas depends entirely on its methane (CH_4) fraction, as carbon dioxide acts as an inert diluent. Rather than calculating the moles of methane independently, this fraction is integrated directly into the mass-based Calorific Value (CV_{mass}) of the specific biogas mixture being used. For these baseline calculations, the CV_{mass} of the biogas is established at 21.67 MJ/kg.

3.5.4 Stored Chemical Energy Estimation

The total stored energy (E_{total}) inside the vessel is the product of the total mass of the gas and its calorific value. Substituting the density equation derived above, the total energy can be expressed as:

$$E_{total} = \left(\frac{P \cdot M}{R \cdot T} \right) \times V_{cylinder} \times CV_{mass} \quad (4)$$

3.5.5 Volumetric Energy Density Relation

The volumetric energy density (E_v), defining the energy per cubic meter of vessel space, is simply the density multiplied by the calorific value:

$$E_v = \rho \cdot CV_{mass} = \left(\frac{P \cdot M}{R \cdot T} \right) \cdot CV_{mass} \quad (5)$$

3.5.6 Pressure-Volume Tradeoff Analysis

Using the derived equations, a baseline tradeoff analysis was conducted assuming the biogas is compressed to a moderate pressure of 20 bar (2,000,000 Pa) at an ambient temperature of 300 K, within a vessel identical in size to the LPG benchmark (0.0333 m³). Unlike liquefied gases, a compressed permanent gas expands to fill the entire container. Therefore, the 86% fill factor constraint used for LPG is discarded, and 100% of the 0.0333 m³ volume is utilized.

Applying the parameters to the energy equation:

$$\begin{aligned} E_{total} &= \left(\frac{2,000,000 \times 0.027}{8.314 \times 300} \right) \times 0.0333 \times 21.67 \\ E_{total} &= 21.65 \text{ kg/m}^3 \times 0.0333 \text{ m}^3 \times 21.67 \text{ MJ/kg} \\ E_{total} &\approx 15.62 \text{ MJ} \end{aligned}$$

3.5.7 Implications for Vessel Design

The baseline calculation reveals a severe thermodynamic limitation. At a working pressure of 20 bar, a 33.3 L cylinder holds only 15.62 MJ of energy. Comparing this to the actual energy of the LPG cylinder (744.588 MJ), the 20-bar biogas system yields only about 2.1% of the benchmark energy. This falls drastically short of the established 25% to 35% target required for weekly refill practicality.

This mathematical conclusion dictates a critical pivot in the vessel's design parameters. Relying purely on basic isochoric compression at moderate pressures is inadequate. To achieve the energy density required for a viable decentralized system, the project must

either drastically increase the design pressure, which necessitates thick-walled heavy cylinders that destroy portability, significantly increase the physical volume, which again ruins portability, or adopt advanced storage techniques such as Adsorbed Biogas (ABG) utilizing high-surface-area biochar to artificially increase the gas density at 20 bar.

4 Manual Optimization Attempts

4.1 Manual Design Exploration

The initial approach to optimizing the biogas storage vessel focused purely on Method 1, which involved material variation. The goal was to maintain a fixed high target pressure, tested at 200 bar to maximize raw gas energy density, and a fixed internal volume, while cycling through different construction materials to find a viable shell thickness and weight. To do this, the design exploration relied strictly on the allowable stress values sourced from the ASME Boiler and Pressure Vessel Code (BPVC) Section II, Part D, Table 1A for ferrous materials. By keeping pressure and geometry constant, the objective was to determine if utilizing higher grade steel or non-ferrous alloys could yield a portable, practical cylinder.

4.2 Practical Difficulties Encountered

The primary difficulty emerged immediately during the material screening process, specifically the inverse relationship between material affordability and structural weight at high pressures.

A shortlist of ferrous materials was compiled and calculated for required wall thickness (t) and the thickness-to-radius ratio (t/R) at 200 bar:

- **IS 2062:** With an allowable stress of 120 MPa, the required thickness is approximately 70 mm ($t/R = 0.23$). This is very thick and entirely impractical for a portable vessel.
- **SA-299 Gr B:** With an allowable stress of 158 MPa, the required thickness is approximately 50 mm ($t/R = 0.17$). This shows improved strength but is still far too heavy.
- **SA-517:** With an allowable stress of 227 MPa, the required thickness is approximately 35 mm ($t/R = 0.12$). This represents a reasonable thickness using high-strength steel.
- **SA-723 (Forging):** With an allowable stress of 374 MPa, the required thickness is approximately 20 mm ($t/R = 0.07$). This is an excellent thickness but highly expensive.

To bypass the weight issues of steel, non-ferrous options were evaluated. While aluminum alloys offer a much lower baseline density, this advantage is completely offset by their significantly lower allowable stress values. To hold 200 bar, an aluminum vessel would require extremely thick walls, negating the density benefits. Other advanced non-ferrous

alloys possess better allowable stress limits. However, their physical density is comparable to the shortlisted ferrous materials, and they introduce severe manufacturing constraints, such as strict thickness range limitations, alongside prohibitively high costs.

4.3 Impracticalities Observed in Preliminary Designs

The conclusion from the preliminary manual designs was that the material variation method inherently failed due to the excessively high weight required to contain the 200 bar pressure. While the best allowable stress-to-density ratio was found in the ultra-high-grade steel (SA-723), it is a forged material. Relying on expensive forgings or exotic alloys entirely defeats the project’s core objective of creating a low-cost, accessible decentralized energy storage system. Attempting to brute-force a high-pressure vessel by simply swapping materials proved economically and practically unviable.

4.4 Results

The failure of the material-swapping approach led to a critical engineering pivot toward Method 2, which focuses on varying pressures and dimensions.

Rather than fixing the pressure at 200 bar and searching for a miracle material, this second method treats pressure and internal geometry as flexible variables. By deliberately lowering the target pressure, standard, cheap, and easily manufactured materials, like IS 2062 or SA-299, can be utilized without the wall thickness ballooning to impractical sizes. While lowering the pressure reduces the raw volumetric energy density, this deficit can be mitigated by slightly altering the vessel’s dimensions within the permitted LPG benchmark envelope.

However, attempting to execute Method 2 manually revealed a new set of challenges. Finding the precise thermodynamic and structural optimum, where pressure is high enough to meet energy targets but low enough to use standard steel, required recalculating parameters across dozens of configurations. The sheer volume of manual iterations needed to balance pressure, volume, thickness, weight, and material cost simultaneously became overwhelming and highly prone to error. It became evident that manual calculation is entirely inadequate for solving this multi-variable problem efficiently. This realization directly motivated the abandonment of hand calculations in favor of an algorithmic, computational optimization framework, the methodology and execution of which are detailed in the subsequent sections of this report.

5 Algorithmic Optimization Attempt

5.1 Need for Computational Optimization

Initial attempts at optimization were carried out manually through iterative calculations using ASME pressure vessel relations and thermodynamic storage equations. However, due to the strongly coupled nature of the governing equations, it rapidly became difficult to simultaneously track the effects of geometry, pressure, material selection, stored energy and vessel mass.

An increase in storage pressure increased the achievable stored energy, but simultaneously increased the required shell and head thicknesses according to ASME stress relations [8], thereby increasing overall vessel mass. Similarly, increasing vessel dimensions increased the internal storage volume and stored energy, but also increased material usage and fabrication requirements. Since these relationships were nonlinear and mutually coupled, repeated manual iteration proved increasingly inefficient and impractical.

Consequently, a MATLAB based optimization framework was developed to automate the design exploration process and systematically identify candidate geometries satisfying the required engineering constraints.

The optimized geometries obtained from the MATLAB solver were subsequently reconstructed within Autodesk Fusion 360 for geometric verification and preliminary CAD validation. Preliminary attempts were also made to incorporate computational FEA driven validation directly into the optimization loop itself, inspired by integrated optimization methodologies available in the literature [12]. However, due to software instability, debugging complexity, and the anticipated computational expense associated with repeated meshing and solving operations during optimization, this approach was abandoned in favour of analytical ASME based optimization followed by post-optimization CAD verification of only the final optimized geometry.

The optimization framework therefore remained fundamentally analytical in nature and relied primarily on established ASME pressure vessel design relations [8] rather than computationally intensive FEA-coupled optimization procedures.

Furthermore, AI-based optimization models were intentionally not employed. The pressure vessel optimization problem is already a well studied constrained optimization benchmark problem with established analytical behaviour and known numerical solution strategies [13]. The use of significantly more computationally expensive AI-based optimization methods was therefore considered unnecessary for the present scope of work. Additionally, computational efficiency itself was treated as a relevant engineering resource in keeping with sustainability-oriented engineering philosophy.

5.2 Definition of the Design Problem

The design objective was to determine an optimal cylindrical pressure vessel geometry with semi-ellipsoidal heads capable of storing compressed biogas while minimizing structural mass and simultaneously maximizing stored energy.

The selected vessel geometry consisted of:

- A cylindrical shell section
- Two ASME 2:1 semi-ellipsoidal heads

The principal design variables were:

$$x = [r, L, t_s, t_h, P]$$

where:

- r = internal vessel radius
- L = cylindrical shell length
- t_s = shell thickness
- t_h = head thickness
- P = internal storage pressure

The internal volume of the vessel was expressed as:

$$V = \pi r^2 L + \frac{4\pi r^2 h}{3}$$

For ASME 2:1 semi-ellipsoidal heads:

$$h = \frac{r}{2}$$

thus:

$$V = \pi r^2 L + \frac{2\pi r^3}{3}$$

The stored biogas energy relation derived earlier was used in the form:

$$E = 23.4 P_{bar} V$$

where:

- E = stored energy in MJ

- P_{bar} = absolute storage pressure in bar
- V = vessel internal volume in m^3

This relation directly coupled the thermodynamic storage objective with the vessel geometry and pressure selection.

5.3 Research References and Methodology Inspiration

The optimization workflow developed in this work was inspired by classical pressure vessel optimization benchmark studies available in the literature. In particular, the widely used pressure vessel design benchmark problem described by Yang et al. [13] provided significant insight into the structure of constrained vessel optimization problems and the formulation of coupled geometric and stress-based constraints.

Similarly, the work of Patel [11] and the later integrated optimization framework proposed by Zhou et al. [12] demonstrated practical approaches for combining computational optimization methods with pressure vessel structural design methodologies.

These studies demonstrated the suitability of constrained nonlinear optimization approaches for pressure vessel design problems involving competing objectives such as material usage reduction and pressure containment requirements.

The methodology adopted in the present work differed from the classical benchmark formulations primarily in the inclusion of:

- Thermodynamic energy storage considerations
- Variable internal storage pressure
- Material property variation
- Explicit ASME-based vessel thickness calculations

The literature review also motivated the use of analytical optimization methods over brute-force or AI-driven approaches due to the relatively well characterized nature of the design space.

5.4 Selection of Optimization Variables

The selected optimization variables were chosen to directly influence both the structural integrity and storage capability of the vessel.

The variables optimized were:

$$x = [r, L, t_s, t_h, P]$$

The internal radius r and shell length L governed the overall storage volume of the vessel. The shell thickness t_s and head thickness t_h governed the structural capacity of the vessel to withstand internal pressure according to ASME design equations [8]. The internal pressure P directly influenced both the stored energy and the required wall thicknesses.

The use of pressure as an optimization variable significantly increased the coupling between the design variables, since increasing pressure simultaneously:

- Increased stored energy
- Increased required shell thickness
- Increased required head thickness
- Increased total vessel mass

This coupling contributed substantially to the nonlinear nature of the optimization problem.

5.5 Constraint Formulation

The optimization process was constrained using both structural and geometric considerations.

The shell thickness constraint was derived from ASME pressure vessel relations [8]:

$$t_s \geq \frac{Pr}{SE - 0.6P}$$

The semi-ellipsoidal head thickness constraint was similarly expressed as:

$$t_h \geq \frac{PD}{2SE - 0.2P}$$

where:

- P = internal pressure
- S = allowable stress for material
- E = weld efficiency
- D = internal diameter

Stored energy limits were imposed as:

$$130 \leq E \leq 200$$

Geometric constraints were also imposed to prevent unrealistic vessel proportions:

$$1 \leq \frac{L}{D} \leq 6$$

Volume constraints were imposed to maintain practical storage capacity:

$$0.025 \leq V \leq 0.035$$

These constraints collectively restricted the feasible design space to physically manufacturable and structurally valid vessel configurations.

5.6 Objective Function Formulation

The optimization problem was formulated as a weighted multi-objective optimization problem involving:

- Minimization of vessel mass
- Maximization of stored energy

The final objective function used was:

$$f(x) = w_M \frac{M}{M_{ref}} + w_E \left(\frac{E - E_{target}}{\Delta E} \right)^2$$

where:

- M = vessel mass
- M_{ref} = reference mass scaling term
- E = stored energy
- E_{target} = desired energy target
- ΔE = energy normalization range
- w_M, w_E = weighting coefficients

Considerable effort was required to balance the relative influence of the mass minimization and energy maximization terms. Early formulations resulted in the optimizer collapsing toward minimum feasible energy values due to excessive dominance of the mass minimization term. Multiple normalization and scaling strategies were therefore explored before arriving at a formulation which produced physically meaningful trade-offs between storage capability and structural weight.

This balancing process proved to be one of the most challenging aspects of the optimization implementation and highlighted the sensitivity of constrained nonlinear optimization problems to objective function scaling [13].

5.7 MATLAB-Based Optimization Framework

The optimization framework was implemented entirely within MATLAB using the `fmincon()` constrained nonlinear optimization solver.

The computational workflow consisted of:

1. Loading material properties from the prepared dataset
2. Iteratively solving the optimization problem for each material
3. Applying ASME thickness constraints
4. Evaluating energy storage capability
5. Computing vessel mass
6. Exporting optimized candidate designs

The framework was modularized into separate scripts responsible for:

- Geometry calculations
- Mass calculations
- Energy calculations
- Constraint evaluation
- Optimization execution

This modular structure simplified debugging and enabled rapid modification of optimization formulations during development.

5.8 Material Dataset and Property Mapping

The optimization framework required both allowable stress values and density values for candidate pressure vessel materials.

Allowable stress data was obtained from ASME material tables [7], while density values were assembled separately through manually curated datasets based on standard material nomenclature and engineering material references.

The final material dataset was stored in CSV format with entries of the form:

[Material Name,Material Type Allowable Stress, Density]

Material properties were mapped using standardized ASME material naming conventions[7]. 134 candidate ferrous materials were included in the final dataset used during optimization.

5.9 Post-Optimization CAD and FEA Validation

Following completion of the MATLAB based optimization process, the optimized vessel geometries were reconstructed within Autodesk Fusion 360 for post-optimization validation and geometric verification purposes.

The reconstructed CAD models consisted of:

- Cylindrical shell sections
- Semi-ellipsoidal heads
- Hollow pressure vessel geometries using the computed shell and head thicknesses

The primary objective of this stage was not iterative FEA-driven optimization, but rather verification of:

- Geometric consistency of the optimized vessel
- Correctness of shell and head thickness calculations
- Physical plausibility of the optimized geometry
- Agreement between analytical and CAD-derived mass values

The optimized vessel dimensions obtained from MATLAB were manually reconstructed using the calculated:

- Internal radius
- Cylindrical shell length
- Shell thickness
- Semi-ellipsoidal head depth
- Head thickness

During the reconstruction process, discrepancies were initially observed between analytically predicted mass values and CAD-derived mass values. Investigation revealed that these discrepancies originated from shell thickness interpretation errors within the CAD geometry construction process. After correction of the geometric construction methodology, the CAD-derived mass values showed close agreement with the analytically predicted values, with deviations below approximately 2%.

This agreement provided confidence that the implemented geometry generation and mass calculation procedures were physically consistent and correctly represented the intended vessel geometry.

Preliminary static structural validation was also performed through pressure loading studies within the CAD environment. These studies were intended only as qualitative post-optimization verification tools rather than primary optimization drivers. Consequently, the final optimization framework relied primarily on analytical ASME pressure vessel relations [8] for structural sizing, while CAD reconstruction and limited post-optimization FEA validation were retained as supplementary verification procedures. The FEA validation does conclude that the designed that was optimized finally was quite reliable with a high safety factor and a mostly green Von Mises stress coloration.[The FEA results are attached as a HTML file and the CAD made manually is also present as an FDX file in the outputs folder of the repository.]

5.10 Proposed Improvements to the Optimization Pipeline

Although the developed MATLAB based optimization framework successfully generated physically reasonable candidate vessel geometries, several possible improvements to the computational pipeline were identified during the course of the work.

The first proposed improvement involved extension of the post-processing stage through automated extraction of the optimal candidate geometry from the optimization results table. This would allow the best performing design to be directly transferred into downstream CAD and validation workflows without requiring manual intervention.

A second proposed improvement involved computational generation of axisymmetric vessel geometries directly within MATLAB. Instead of relying entirely on external CAD reconstruction, simplified vessel geometries consisting of:

- Quarter cylindrical shell sections
- Quarter semi-ellipsoidal head profiles

could be generated programmatically and reconstructed through axisymmetric symmetry operations. Such an approach would significantly reduce geometric modeling effort while preserving the dominant structural characteristics of the pressure vessel.

The most significant proposed extension to the optimization framework involved integration of FEATool based structural validation directly within MATLAB. Since FEATool operates natively within the MATLAB environment, it presents the possibility of substantially more efficient coupling between:

Optimization → Geometry Generation → Meshing → Structural Validation

compared to workflows relying on external CAD/FEA software communication.

A FEATool based workflow could potentially allow:

- Automated geometry generation
- Axisymmetric meshing
- Internal pressure loading
- Stress evaluation
- Direct feedback into the optimization loop

within a single computational environment.

Such an approach would significantly reduce software interoperability issues while potentially enabling limited in-loop FEA validation with acceptable computational efficiency.

However, due to project scope limitations, software stability concerns, and the additional complexity associated with robust automated meshing and geometry handling, this extension was not fully implemented within the present work.

Nevertheless, the proposed workflow remains a promising direction for future development of a more tightly integrated computational pressure vessel optimization framework.

6 Conclusion

This report details the evolution of a decentralized biogas storage vessel, transitioning from conventional high-pressure and CAES concepts to a structurally optimized, moderate-pressure solution. Initial manual design explorations revealed a strict thermodynamic limitation: storing standard compressed biogas at safe, portable pressures yields inadequate energy density for practical domestic use, while high-pressure solutions demand unviably thick and heavy steel walls.

To overcome these bottlenecks, the project established two critical engineering pathways:

- **Algorithmic Optimization:** An automated MATLAB framework was developed to replace inefficient manual calculations. This computational approach successfully balances ASME structural constraints, material mass, and internal volume to mathematically identify the exact geometric and pressure optimums for the vessel.
- **Adsorbed Biogas (ABG) Integration:** To artificially increase gas density at a manageable 20 bar pressure, the integration of high-surface-area activated biochar was identified as a necessary mechanism. This physical adsorption technique significantly enhances storage capacity without requiring heavier industrial-grade steel.

Ultimately, the synergy of computational optimization and ABG technology fulfills the primary motivation of this project. By engineering a storage capacity and portability profile comparable to a standard LPG cylinder, this proposed vessel provides a safe, scalable, and low-cost energy solution. It holds the potential to empower rural households to effectively store and utilize locally generated, renewable biomethane for daily cooking and domestic needs, offering a pathway to circumvent the reliance on expensive industrial compression infrastructure.

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